

Statistical Methods I: Linear Model

Lec 01: Introduction to Simple Linear Regression

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Section 1

Introduction

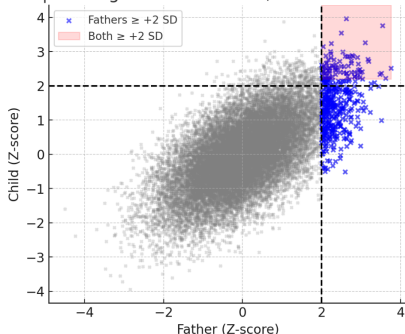
History of regression

- 1700: Newton wrote normal equations for least squares, several other researchers did similar calculations
- 1805: Legendre published method of least squares
- 1809: Gauss claimed to have used least squares since 1795
- 1821: Gauss published his method of least squares, introduced maximum likelihood and normal distribution, discussed optimality
- 1886: Galton published “Regression towards mediocrity in hereditary stature”

Regression to the mean

- Galton observed that extreme values on one measurement tend to be associated with less extreme values on another measurement.
- This phenomenon is now known as “regression to the mean”.

$\rho = 0.6$ | Among fathers $\geq +2$ SD, children $\geq +2$ SD = 24.2%



After Galton

- 1896: Pearson (Galton's student) developed correlation coefficient
- 1901: Yule (Pearson's student) developed multiple regression
- 1920s: Fisher (Pearson's student) developed analysis of variance (ANOVA), F-tests

Why Do We Still Need to Learn Linear Regression?

- **Foundation of statistical modeling**

- Still widely used today
- Basis for many modern methods (GLM, mixed models, ridge, Lasso, deep learning layers, ...)

- **Interpretability**

- Coefficients have clear meaning: expected change in outcome per unit increase in predictor
- Provides intuition about relationships between variables

- **Practical importance**

- Still used daily in almost all areas: medicine, economics, engineering, and social sciences

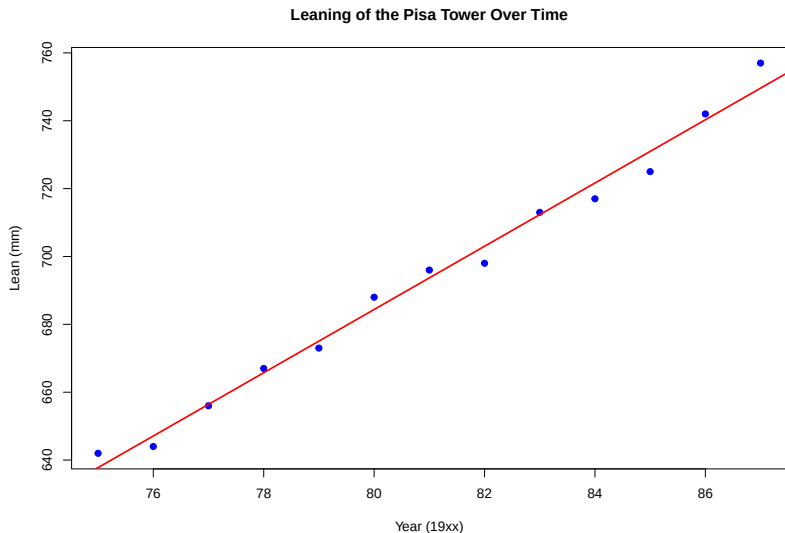
Example: Leaning tower of Pisa

- Response variable: Lean of the tower (in mm)
- Predictor variable: Year (19xx)



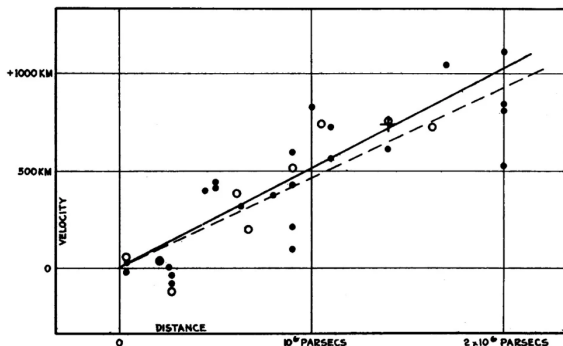
OBS	YEAR	LEAN
1	75	642
2	76	644
3	77	656
4	78	667
5	79	673
6	80	688
7	81	696
8	82	698
9	83	713
10	84	717
11	85	725
12	86	742
13	87	757
14	100	.

Example: Leaning tower of Pisa



Example: Hubble's law (1929)

- Galaxies are moving away from us
- the farther away a galaxy is, the faster it is moving away



Section 2

SLR

Simple Linear Regression

- In a Simple Linear Regression Model, a single predictor (independent/explanatory variable) is used to predict a response/dependent variable

Simple Linear Regression

- A simple linear regression line describes a linear relationship between ___ and ___:

$$E(y|x) = \beta_0 + \beta_1 x$$

- The simple linear regression model

$$y = \beta_0 + \beta_1 x + \epsilon$$

where ϵ is random error.

Notation

- For the individual i :

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n$$

- β_0, β_1 : regression coefficients (unknown parameters)
 - β_0 : intercept
 - β_1 : slop
- y_i : observed response value of individual i
- x_i : observed predictor value of individual i
- ϵ_i : random error of individual i

Assumptions: Basic (a)

- Basic assumptions

- 1 Linearity: The relationship between the predictor and the mean of the response is linear.

$$E(y_i|x) = \beta_0 + \beta_1 x_i$$

Equivalently,,

$$E(\epsilon) = 0$$

- 2 Uncorrelated errors: The errors are uncorrelated.

$$\text{Cov}(\epsilon_i, \epsilon_j) = 0, \quad i \neq j$$

- 3 Homoscedasticity: The errors have constant variance at all levels of the predictor.

$$\text{Var}(\epsilon_i) = \sigma^2, i = 1, \dots, n$$

Assumptions: Stronger (b)

- Stronger assumptions
 - 1 Linearity
 - 2 Independence: The errors are independent.
 - 3 Normality: The errors are normally distributed.
- The above assumptions are equivalent to

$$\epsilon_1, \dots, \epsilon_n \stackrel{iid}{\sim} N(0, \sigma^2)$$

Implications of Assumptions (a)

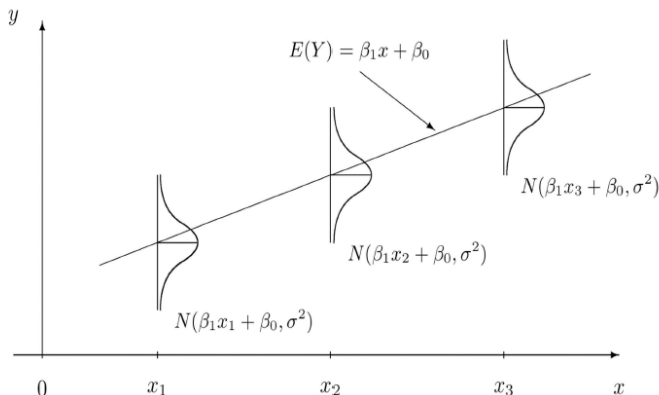
Regression Model: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n$

- For a given x_i , under the basic assumptions (a) or (b), we have
 - y_i is a random variable
 - $E(y_i|x_i) = \beta_0 + \beta_1 x_i$
 - $Var(y_i|x_i) = \sigma^2$
 - $Cov(y_i, y_j|x_i, x_j) = 0, i \neq j$, uncorrelated, but not necessarily independent

Implications of Assumptions (b)

Regression Model: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n$

- For a given x_i , under the basic assumptions (b), we have
 - $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$
 - $y_1, \dots, y_n$ are independent



Section 3

Estimate β_0, β_1

Estimate coefficients

- Predicted / Fitted values $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.
- Observed errors (residuals) $e_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$.
- Sum of Squared Errors (SSE), also known as Residual Sum of Squares (RSS):

$$RSS = SSE = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

- The Least Squares Estimation (LSE) criterion chooses $\hat{\beta}_0, \hat{\beta}_1$ that minimizes RSS (SSE).

Normal Equations

- The normal equations are obtained by setting the partial derivatives of RSS with respect to $\hat{\beta}_0$ and $\hat{\beta}_1$ to zero:

$$\frac{\partial RSS}{\partial \hat{\beta}_0} = -2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\frac{\partial RSS}{\partial \hat{\beta}_1} = -2 \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

- Solving these equations gives the least squares estimates.

Least Squares Estimates

- The least squares estimates for the coefficients are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ and $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ are the sample means of x and y , respectively.

$\hat{\beta}_0, \hat{\beta}_1$ are called least squares estimators.

Sum of Squares in Simple Linear Regression

- $S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2$, total sum of Squares of y
- $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$, total sum of Squares of x
- $S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$, covariance sum of Squares.

The least squares estimates can be expressed as:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

It can be shown that $SSE = S_{yy} - \hat{\beta}_1 S_{xy} = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$.

Properties of Fitted Values and Residuals

- The sum of the residuals is zero: $\sum_{i=1}^n e_i = 0$.
- The least squares regression line passes through the point $(\bar{x}, \bar{y})$ because $\hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y}$ (by the normal equation).
- The sum of the observed values = the sum of the fitted values:

$$\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i$$

- The sum of the residuals weighted by the predictor values or predicted values is zero:

$$\sum_{i=1}^n x_i e_i = 0, \sum_{i=1}^n \hat{y}_i e_i = 0$$

Section 4

Estimate Variance

Estimate σ^2

- An intuitive estimate of σ^2 might be:

$$s^2 = \frac{1}{n} \sum_{i=1}^n e_i^2 = \frac{SSE}{n}$$

- However, this estimate is biased because the residuals e_i are not independent and have less variability than the true errors ϵ_i .
- An unbiased estimate of σ^2 is:

$$s^2 = \frac{1}{n-2} \sum_{i=1}^n e_i^2 = \frac{SSE}{n-2} = MSE$$

Section 5

Coefficient of Determination

Sum of Squares Decomposition

- Total Sum of Squares (SST):

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

- Regression Sum of Squares (SSR):

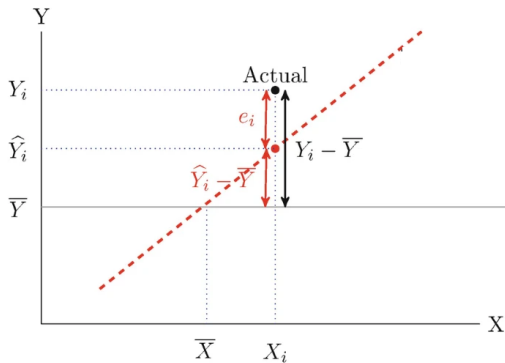
$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

The variation explained by the predictor (regression model).

- Error Sum of Squares (SSE):

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Sums of Squares



It can be shown that:

$$SST = SSR + SSE$$

Coefficient of Determination R^2

- The coefficient of determination R^2 is defined as:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

- R^2 represents the proportion of variance in the response variable that is explained by the predictor variable.

Correlation

- The correlation coefficient, ρ , provides a measure of the strength and direction of a linear relationship between two variables, x and y .
- The sample correlation coefficient r between x and y is given by:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

- It can be shown that in simple linear regression, the coefficient of determination R^2 is equal to the square of the sample correlation coefficient r :

Summary

- LSE of β_0, β_1 :

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

- Variance estimate:

$$s^2 = \frac{SSE}{n-2} = \frac{1}{n-2} \left(S_{yy} - \frac{S_{xy}^2}{S_{xx}} \right)$$

- Coefficient of determination:

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST} = \frac{S_{xy}^2}{S_{xx} S_{yy}} = r^2$$

Useful Formulas

- Useful formulas for calculating S_{XX}, S_{YY}, S_{XY} :

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$$

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2$$

$$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}$$

- Sufficient statistics for conducting regression analysis :
 $n, \sum x_i, \sum y_i, \sum x_i^2, \sum y_i^2, \sum x_i y_i$

Section 6

Example

Example: Battery data

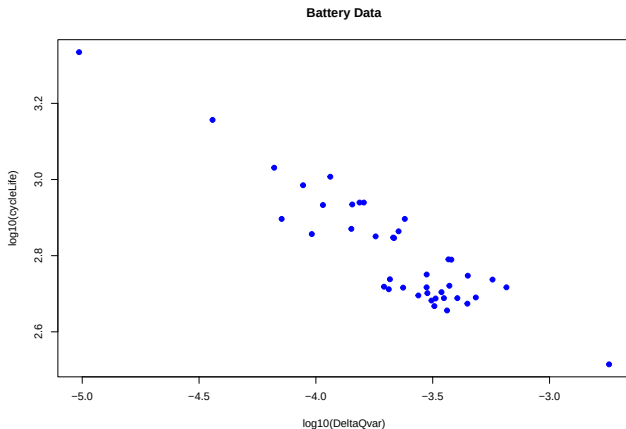
Background: Predicting cycle life of lithium-ion batteries from initial cycles

Source: MathWorks Predictive Maintenance Toolbox

- x : difference in discharge capacity as a function of voltage between 100th and 10th cycle
- y : cycle life (number of cycles until the battery capacity falls below 80% of its initial capacity)
- $\log_{10}$ both variables to get a linear relationship

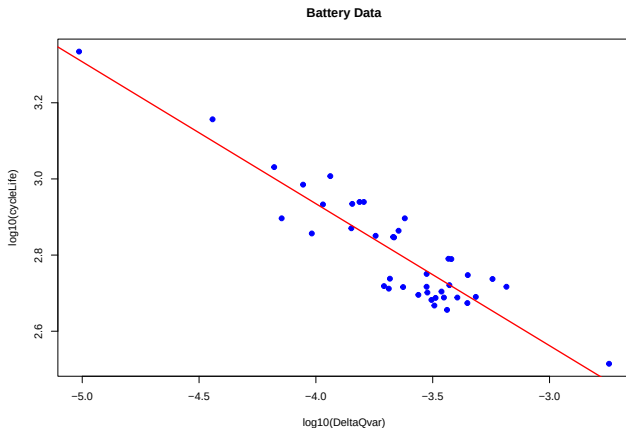
Example: Battery data

Scatter plot:



Example: Battery data

- Fit a linear regression



Example: Battery data

- Summary of the fitted model

```
> summary(fit)
```

```
call:
```

```
lm(formula = log10(cycleLife) ~ log10(DeltaQvar), data = battery)
```

```
Residuals:
```

Min	1Q	Median	3Q	Max
-0.107697	-0.056378	0.009384	0.056817	0.103507

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.44364	0.09990	14.45	<2e-16 ***
log10(DeltaQvar)	-0.37281	0.02718	-13.72	<2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.06424 on 39 degrees of freedom
```

```
Multiple R-squared:  0.8283,    Adjusted R-squared:  0.8239
```

```
F-statistic: 188.2 on 1 and 39 DF,  p-value: < 2.2e-16
```

Example: Battery data

Do the calculations “manually”:

- sufficient statistics:

```
## [1] 41
```

```
## [1] -149.953
```

```
## [1] 115.0934
```

```
## [1] 554.024
```

```
## [1] 324.0225
```

```
## [1] -423.0244
```

```
## [1] -3.65739
```

```
## [1] 2.807155
```

```
## [1] 5.587388
```

```
## [1] 0.9375112
```

```
## [1] -2.083033
```

How to Calculate Simple Linear Regression Estimates

Given summary statistics

- $n = 41$
- $\sum x_i = -149.953, \sum y_i = 115.0934$
- $\sum x_i^2 = 554.024, \sum y_i^2 = 324.0225$
- $\sum x_i y_i = -423.0244$

Step 1. Compute sufficient statistics

$$\bar{x} = \frac{-149.953}{41} \approx -3.66, \quad \bar{y} = \frac{115.0934}{41} \approx 2.81$$

$$S_{xx} = 554.024 - \frac{(-149.953)^2}{41} \approx 5.59$$

$$S_{yy} = 324.0225 - \frac{(115.0934)^2}{41} \approx 0.94$$

$$S_{xy} = -423.0244 - \frac{(-149.953)(115.0934)}{41} \approx -2.08$$

Step 2. Regression coefficients

$$\hat{\beta}_1 = \frac{s_{xy}}{s_{xx}} \approx \frac{-2.08}{5.59} \approx -0.37$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \approx 2.81 - (-0.37)(-3.66) \approx 1.46$$

Step 3. Error variance

$$\hat{\sigma}^2 = \frac{1}{n-2} (S_{yy} - \hat{\beta}_1 S_{xy}) = \frac{1}{39} (0.94 - (-0.37)(-2.08)) \approx \frac{0.17}{39}$$

$$\hat{\sigma} \approx \sqrt{0.0044} \approx 0.066;$$

Interpretation of coefficients

- $\hat{\beta}_1 \approx -0.37$: For a one unit increase in $\log_{10}(\text{DeltaQvar})$, the expected change in $\log_{10}(\text{cycleLife})$ is a decrease of approximately 0.37 units.
- $\hat{\beta}_0 \approx 1.46$:

Coefficient of Determination

- $R^2 = \frac{S_{xy}^2}{S_{xx}S_{yy}} \approx \frac{(-2.08)^2}{5.59 \times 0.94} \approx 0.82$
- About 82% of the variability in $\log_{10}(\text{cycleLife})$ can be explained by the linear relationship with $\log_{10}(\text{DeltaQvar})$.
- $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \approx \frac{-2.08}{\sqrt{5.59 \times 0.94}} \approx -0.91$.
- The strong negative correlation (-0.91) indicates a strong inverse linear relationship between $\log_{10}(\text{DeltaQvar})$ and $\log_{10}(\text{cycleLife})$.
- We will learn more about how to interpret the coefficients when transformation is used in later lectures.